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# =====
# STUDENT PERFORMANCE ANALYSIS MINI PROJECT
# =====

# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# -----
# Dataset
# -----

data = {
    "Student_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Asha", "Rahul", "Priya", "Kiran", "Sneha",
             "Ravi", "Anjali", "Vikram", "Pooja", "Arjun"],
    "Maths": [85, 70, 92, 60, 78, 88, 95, 67, 80, 73],
    "Science": [90, 65, 89, 58, 82, 91, 96, 72, 84, 75],
    "English": [88, 75, 91, 62, 80, 85, 94, 70, 79, 77],
    "Attendance (%)": [95, 80, 98, 70, 85, 92, 99, 78, 88, 83]
}

# -----
# Create DataFrame
# -----

df = pd.DataFrame(data)

print("Student Dataset")
print(df)

# -----
# Calculate Total Marks
# -----

df["Total"] = df["Maths"] + df["Science"] + df["English"]

# -----
# Calculate Average Marks
# -----

df["Average"] = df["Total"] / 3

# -----
# Assign Grades
# -----

def grade(avg):
    if avg >= 90:
        return "A+"

    elif avg >= 80:
        return "A"

    elif avg >= 70:
        return "B"

    elif avg >= 60:
        return "C"

    else:
        return "Fail"

df["Grade"] = df["Average"].apply(grade)

# -----
# Display Updated Dataset
# -----
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print("\nStudent Performance Report")
print(df)

# -----
# Top Performer
# -----

top_student = df.loc[df["Average"].idxmax()]

print("\nTop Performer")
print(top_student)

# -----
# Subject-wise Average
# -----

print("\nSubject-wise Average Marks")

print("Maths Average :", np.mean(df["Maths"]))
print("Science Average :", np.mean(df["Science"]))
print("English Average :", np.mean(df["English"]))

# -----
# Attendance Analysis
# -----

high_attendance = df[df["Attendance (%)"] > 90]

print("\nStudents with Attendance Above 90%")
print(high_attendance[["Name", "Attendance (%)"]])

# =====
# DATA VISUALIZATION
# =====

# -----
# Bar Chart - Student Average Marks
# -----

plt.figure(figsize=(10,5))

plt.bar(df["Name"], df["Average"])

plt.title("Student Average Marks")

plt.xlabel("Student Names")

plt.ylabel("Average Marks")

plt.xticks(rotation=45)

plt.show()

# -----
# Pie Chart - Grade Distribution
# -----

grade_count = df["Grade"].value_counts()

plt.figure(figsize=(6,6))

plt.pie(grade_count,
        labels=grade_count.index,
        autopct='%1.1f%%')

plt.title("Grade Distribution")

plt.show()

# -----
# Line Chart - Attendance
# -----

plt.figure(figsize=(10,5))

plt.plot(df["Name"],
         df["Attendance (%)"],

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marker='o')

plt.title("Attendance Analysis")

plt.xlabel("Students")

plt.ylabel("Attendance Percentage")

plt.grid(True)

plt.show()

# -----
# Save Report to CSV
# -----

df.to_csv("student_performance_report.csv", index=False)

print("\nCSV Report Saved Successfully")
```

